I've spent way too much time toying with my TI-84 calculator during math classes — and not actually doing work I am supposed to. I think I tried basically every feature on this calculator without consulting the manual, haha.

The model I use — CE Plus Python — was released in 2015. It's wildly expensive for what can pretty much be done in Desmos, but oh well.

CalcLang is a Toki Pona-inspired minimalistic conlang fully expressible on a TI-84 calculator. I attempted to make the meanings of operators roughly semantically similar to their mathematical meaning. All grammatical sentences are valid mathematical operations.

Finally I can write poetry during math class! It is particularly fun for constrained writing exercises. A fun exercise is to constrain evaluated results in a pattern such as having all sentences evaluate to 1.

Grammar

CalcLang uses SVO order. Subject, predicate, and direct object are separated by the particle '+'. CalcLang is head-final, meaning a modifier is followed by the modified noun or verb.

```
1  +   6    +   2
1S SEP love SEP 2S
```

Its minimal vocabulary is represented by numbers. Content words may function as a noun, verb, or modifier with zero derivation.

'*' separates a word of the same part of speech preceding it, a function roughly analogous to 'and'.

```
1  *   2  +    6    +   3
1S AND 2S SEP  love SEP 3S
```

'-', when used as a unary function, negates the following word. As a binary function, it indicates locative. Fractions and divisions indicate genitive.

```
1  +   40    -   23      /      40
1S SEP slept LOC surface GEN  sleep
lit: I slept in bed
```

'.' indicates an "essence modifier", which indicates an unalienable quality. They are lifted from aUI, with modifications. Think of the word "necessarily". The same number takes a different meaning as essence modifiers. For example 3 as a regular content word refers to a third person pronoun, but 3 as an essence modifier indicates inalienable humanity.

```
3     /   48
human GEN art
lit: human-made art
```

```
3     .   48
human ESS art
translation: art is necessarily human
```

Exponents act as reference markers. 4^1 means the first thing mentioned in context. 4^2 the second and so on. Negative exponents counts from the last referent. 4^-1 would mean the last thing mentioned. A zero exponent takes on a reflexive meaning: 4^0 would mean “the thing itself”.

On a similar note, the ANS variable—which in TI-84 stores the last evaluation—is a reference marker for the entirety of the last sentence.

```
Mama is baking bread.

49 + 7 + 40

I can smell mom baking the bread.

1 + (37 / 51) + ANS.
```

“0” is a dummy word used to indicate an omitted subject or object.

```
The light was extinguished.
0 + 32 + 31
```

Questions

- Yes-or-no questions are formed in the same way as Toki Pona
- For WH-questions, summation indicates questions, where the x indicates the desired word.

Imperatives / emphasis

- Simply append a factorial for imperatives!

```
Drink this water!
0 + 38 + 4 / 41!
```

- It can also be appended to a specific word for emphasis.

```
The creature was evil!

(28 + 21)!

The creature was the evil one.
```

```
28! + 21 / 4
```

Comparisons

‘log(x)’ means x is in relation to, compared to (Think: Logarithmic scales)
‘$\log_y(x)$’ means x is in relation to y

Cause and effect

Roots mark causation. Higher the degree of the root, the more distant the cause is.

Sample Translation

From *The Very Hungry Caterpillar*:

```
In the light of the moon a little egg lay on a leaf.
(34 / 8.18) + 36 + 33 - (31 / 32)

He started to look for some food.
3 + 42 / 44 + sqrt(38)

On Monday he ate through one apple.
3 + 38 + 27 / 39

But he was still hungry.
log(-ANS) * (3 + 38 / 13)
```